Quantitative Observation-Law Certificates for the High-Temperature Sherrington–Kirkpatrick Gap

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Technical Report Assisted by AI Models
www.integralresearch.net
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Technical Report TR-Ram-OpenAI-722-M429

October 9, 2026

Abstract

We study the observation-law transfer in the September 24, 2026 manuscript *A spectral gap throughout the high-temperature Sherrington–Kirkpatrick phase*. Its ordinary posterior and its posterior reweighted by an arbitrary squared test function are different probability laws. Combining the source's unnormalized ordinary-mean estimate with its square-density estimate gives a sharper drift budget. A sharp elementary affine envelope makes the leading entropy error quadratic in the fixed selection tolerance. We derive explicit finite-parameter Poincare certificates, including a monotone binary-entropy root certificate and a closed lower bound, without replacing the source's function-independent stopping or terminal all-function inequality. A second result transfers these certificates through a uniformly bounded physical log-likelihood residual, with the same reweighting applied to both laws. Exact rational examples demonstrate when the sharper budget is informative; counterexamples explain why covariance positivity or ordinary rarity alone cannot supply the missing gap. The random-matrix, stable-root, finite-recipe and high-probability inputs remain upstream obligations, not consequences of finite computations. Contribution assessment: C2/5, the report sharpens a specific entropy-transfer budget and gives explicit residual certificates using classical inequalities, without a new high-temperature mechanism.

1 Model, clocks, and the claim being studied

Fix $0 < \beta < 1$, put $j = \beta^2$, and let J be a symmetric n by n matrix with zero diagonal and independent upper-triangular entries $J_{ij} \sim N(0, j/n)$. On $\{-1, 1\}^n$ the source uses

$$\mu_h(x) = Z_h^{-1} \exp\{\tfrac{1}{2} x^{\mathsf{T}} J x + h^{\mathsf{T}} x\}.$$

For any full-support probability μ on this cube, write $P_i f = \mathbb{E}_\mu[f \mid x_{-i}]$ and

$$\mathcal{D}_\mu(f) = \sum_{i=1}^{n} \mathbb{E}_\mu (f - P_i f)^2, \qquad L = \sum_i (P_i - I), \qquad \operatorname{gap}(\mu) = \inf_{\operatorname{Var}_\mu f > 0} \frac{\mathcal{D}_\mu(f)}{\operatorname{Var}_\mu f}. \tag{1}$$

This is the continuous-time rate-one-per-site clock. The discrete chain $P = n^{-1} \sum_i P_i = I + L/n$ has gap $\operatorname{gap}(\mu)/n$, not $\operatorname{gap}(\mu)$. For $d_i f = (f(x) - f(x^{(i)}))/(2x_i)$ and $v_i(x) = \operatorname{sech}^2(h_i + (Jx)_i)$,

$$\mathcal{D}_{\mu_h}(f) = \mathbb{E}_{\mu_h} \sum_i v_i (d_i f)^2.$$

The conditional-variance weights cannot be dropped in a physical certificate. Adding a diagonal to J changes the Hamiltonian only by a constant, but changing the normalization of off-diagonal entries changes the parameter j.

The source [1] claims a constant $C_\beta < \infty$ such that, with probability tending to one over J, $\mathrm{Var}_{\mu_0} f \le C_\beta \mathcal{D}_{\mu_0}(f)$ simultaneously for every f. The fixed β, zero external field, typical disorder and all-function quantifiers are part of that statement. The present report does not independently establish its deep probabilistic inputs or turn the result into an arbitrary-field, all-disorder, or log-Sobolev theorem.

2 The exact Gaussian observation interface

Here π is any full-support law on the cube, fixed independently of a standard n-dimensional Brownian motion B. Let $X \sim \pi$, $Y_t = tX + B_t$, and let $\mathcal{F}_t$ be the observation filtration. The ordinary posterior is

$$\pi_t(x) = \mathbb{P}(X = x \mid \mathcal{F}_t) = \frac{\pi(x) e^{Y_t^\mathsf{T} x}}{\sum_z \pi(z) e^{Y_t^\mathsf{T} z}}. \tag{2}$$

The factor $e^{-t\|x\|^2/2} = e^{-tn/2}$ cancels only because cube spins have the same norm. In the SK application $\pi = \mu_0$ and $\pi_t = \mu_{Y_t}$. For fixed g the innovation $I_t = Y_t - \int_0^t \mathbb{E}_{\pi_s} x \, ds$ is Brownian and

$$d\,\mathbb{E}_{\pi_t} g = \mathrm{Cov}_{\pi_t}(g, x)^\mathsf{T} dI_t. \tag{3}$$

These standard finite-state filtering identities are also derived in the source's observation module; they are not new here.

Fix f with $\pi f = 0$, $\pi f^2 = 1$, and put

$$M_t = \pi_t f, \quad N_t = \pi_t f^2, \quad V_t = N_t - M_t^2, \quad d = \mathcal{D}_\pi(f), \quad R_t = \mathcal{D}_{\pi_t}(f)/N_t.$$

All $N_t > 0$. Under the law $\mathbb{P}^f$ obtained by changing the initial law of X to $f^2\pi$, the likelihood on $\mathcal{F}_t$ is N_t, and its posterior is $\nu_t = f^2\pi_t/N_t$. In particular

$$a_t = \mathbb{E}_{\nu_t} x - \mathbb{E}_{\pi_t} x, \qquad dN_t = N_t a_t^\mathsf{T} dI_t, \qquad \|a_t\| \le 2\sqrt{n}. \tag{4}$$

The square-density law is not the ordinary posterior, and its initial spin law need not be an SK Gibbs law with a changed field.

For any bounded stopping time $\tau \le T$, boundedness in (4) justifies the likelihood change and gives

$$H_f := \mathrm{KL}(\mathbb{P}^f|_{\mathcal{F}_\tau} \| \mathbb{P}|_{\mathcal{F}_\tau}) = \frac{1}{2}\mathbb{E}^f \int_0^\tau \|a_t\|^2 \, dt. \tag{5}$$

The exact identity is a filtering/Girsanov identity at fixed finite n; no uniform-in-n Novikov assertion is needed. The Dirichlet contraction needed below is at deterministic time:

$$\mathbb{E}\mathcal{D}_{\pi_t}(f) \le \mathcal{D}_\pi(f) = d, \qquad \mathbb{E}^f R_t = \mathbb{E}[N_t R_t] = \mathbb{E}\mathcal{D}_{\pi_t}(f) \le d. \tag{6}$$

Indeed, condition on x_{-i} and then on observations; the law of total variance says that the average conditional variance decreases under this additional conditioning. Sum over i. Indicators of $t < \tau$ can be inserted on the left of the second inequality by nonnegativity.

3 A quadratic leading entropy budget

Lemma 3.1 (Sharp affine envelope)**.** *If $0 \le a < 2$, $z \ge 0$, and $0 \le b \le \min(2, a + z)$, then*

$$b^2 \le \min\{4, a^2 + (2 + a)z\}. \tag{7}$$

Among affine majorants with intercept a^2, the coefficient $2 + a$ is optimal. If only $\mathbb{E}z \le m$ is known, the sharp universal upper bound is $\mathbb{E}b^2 \le \min\{4, a^2 + (2 + a)m\}$.

Proof. For $b \le a$ there is nothing to show. Otherwise $b^2 - a^2 = (b-a)(b+a) \le z(2+a)$. The constraint $b \le 2$ gives the other bound. At $z = 2-a$, $b = 2$, equality forces any such slope to be at least $(4-a^2)/(2-a) = 2+a$. For $0 \le m \le 2-a$, mix the admissible pairs $(z,b) = (0,a)$ and $(2-a,2)$ with respective probabilities $1 - m/(2-a)$ and $m/(2-a)$. For $m \ge 2-a$, take $z = m$, $b = 2$. These examples prove the moment assertion. They certify sharpness for the scalar constraints, not attainability by a particular SK posterior. □

The source's directional argument contains the stronger ordinary-mean conclusion, in unnormalized Euclidean norm,

$$\|\mathbb{E}_{\pi_t} x - t_*\| \le C_0, \qquad \frac{\|\mathbb{E}_{\nu_t} x - t_*\|}{\sqrt{n}} \le C_w \delta + \frac{C_\delta}{\sqrt{n}}(1 + R_t). \tag{8}$$

Here $t_* = \tanh r$ is the source's unique stable-root magnetization. These bounds are assumed on the stopped stability region with their source quantifiers; they do not follow from covariance positivity. The first is explicitly recorded in `consequences.tex`, lines 184–195; the second is its square-density proposition, lines 202–243. The constants C_0, C_w can be fixed independently of the sufficiently small tolerance δ, while C_δ and the finite diagnostic length may depend on δ.

Thus, before stopping,

$$b_t := \|a_t\|/\sqrt{n} \le a + \frac{B_0 + B_1 R_t}{\sqrt{n}}, \qquad a = C_w\delta, \quad B_0 = C_0 + C_\delta, \quad B_1 = C_\delta. \tag{9}$$

The next result also applies to any independently certified constants a, B_0, B_1 satisfying this input.

Proposition 3.2 (Stopped entropy budget)**.** *Suppose $0 \le a < 2$ and $B_0, B_1 \ge 0$ are deterministic constants, uniform over the test functions and observation paths in the stopped region, and (9) holds for $t < \tau$. Then*

$$H_f \le H_0 + H_1 d, \qquad H_f \le 2nT, \tag{10}$$

$$H_0 = \frac{nTa^2}{2} + \frac{\sqrt{n}\,T(2+a)B_0}{2}, \qquad H_1 = \frac{\sqrt{n}\,T(2+a)B_1}{2}. \tag{11}$$

Proof. Apply Lemma 3.1 to b_t and $z_t = (B_0 + B_1 R_t)/\sqrt{n}$ in (5). The constant terms cost at most T, and $\mathbb{E}^f \int_0^\tau R_t\,dt \le Td$ by (6). The separate bound follows from $\|a_t\|^2 \le 4n$. □

For the source choice, H_0/n has leading term $TC_w^2\delta^2/2$, rather than an order-δ term obtained by $b_t^2 \le 2b_t$. This is an actual downstream improvement of the error budget, not an improvement of the source's matrix or root theorem. Using the sharper ordinary-mean conclusion also avoids paying a second leading $C_w\delta$ for the ordinary law. It does not bound C_δ or justify choosing a vanishing $\delta = \delta_n$ without new uniform information.

4 Finite-parameter Poincare certificates

The hypotheses below isolate every global input used in the transfer. They must hold for one common stopping rule and every nonconstant f, after centering and normalizing as above.

1. A stopping time $\tau \le T$ independent of f, and a bad event $E \in \mathcal{F}_\tau$, satisfy $\mathbb{P}(E) \le q < 1$. On E^c, $\tau = T$ and the terminal law satisfies $\mathrm{Var}_{\pi_T} g \le C_T \mathcal{D}_{\pi_T}(g)$ for every g.
2. Before τ, $\|\mathrm{Cov}_{\pi_t}(f,x)\|^2 \le C_d(V_t + \mathcal{D}_{\pi_t}(f))$, with $C_d \ge 0$ independent of f.
3. The drift bound (9) holds with fixed $0 \le a < 2$ and $B_0, B_1 \ge 0$.

The event in the first condition is under the *ordinary* observation law, not under every square-density law. Write

$$0 < v \le e^{-C_d T}, \quad \kappa = C_T + 1 - v > 0, \quad \ell = \log(1/q), \quad \mathrm{kl}(p,q) = p\log\frac{p}{q} + (1-p)\log\frac{1-p}{1-q}.$$

At endpoints $0 \log 0 = 0$. Assume $0 < q < v \le 1$ in the following theorem.

Theorem 4.1 (Explicit and binary certificates)**.** *Under these hypotheses the unscaled gap satisfies*

$$\operatorname{gap}(\pi) \geq \frac{[v - (H_0 + \log 2)/\ell]_+}{\kappa + H_1/\ell}. \tag{12}$$

If $\operatorname{kl}(v, q) > H_0$*, let* d_* *be the unique root in* $(0, (v-q)/\kappa]$ *of*

$$\operatorname{kl}(v - \kappa d_*, q) = H_0 + H_1 d_*. \tag{13}$$

Then $\operatorname{gap}(\pi) \geq d_*$*. This binary certificate is at least the positive lower bound in* (12)*. If its strict initial condition fails, this test makes no positive assertion.*

Proof. Let $u(t) = \mathbb{E}V_{t\wedge\tau}$. By (3) and Itô's formula, $u'(t) = -\mathbb{E}\mathbf{1}_{\{t<\tau\}}\|\operatorname{Cov}_{\pi_t}(f, x)\|^2$ for almost every t. The directional hypothesis, nonnegativity of V and (6) give $u' \geq -C_d(u + d)$, $u(0) = 1$. Hence

$$u(T) \geq e^{-C_d T} - (1 - e^{-C_d T})d \geq v - (1 - v)d. \tag{14}$$

On the good event use the terminal all-function inequality and $\mathbb{E}\mathcal{D}_{\pi_T}(f) \leq d$. On the bad event $V_\tau \leq N_\tau$, and the stopped likelihood identity gives $\mathbb{E}\mathbf{1}_E N_\tau = \mathbb{P}^f(E) =: p$. Therefore $u(T) \leq C_T d + p$, so $p \geq v - \kappa d$.

Data processing for relative entropy gives $\operatorname{kl}(p, q_0) \leq H_f$ where $q_0 = \mathbb{P}(E) \leq q$. The elementary binary entropy bound gives $p \log(1/q) \leq H_f + \log 2$, including the case $q_0 = 0$ by a limit. Insert (10), combine with $p \geq v - \kappa d$, and solve the linear inequality. This proves (12) for every normalized f without a preliminary assumption $d \leq 1$.

For $0 \leq d < (v-q)/\kappa$, put $r = v - \kappa d > q$. If $q_0 > 0$, monotonicity of binary entropy gives $\operatorname{kl}(r, q) \leq \operatorname{kl}(p, q_0) \leq H_0 + H_1 d$: the function increases in its first argument above q, and decreases in its second argument below its first. If $q_0 = 0$, absolute continuity gives $p = 0$, contradicting $p \geq r > 0$. The function $F(d) = \operatorname{kl}(v - \kappa d, q) - H_0 - H_1 d$ is continuous and strictly decreasing on this interval; in its interior

$$F'(d) = -\kappa \log \frac{(v - \kappa d)(1 - q)}{q(1 - v + \kappa d)} - H_1 < 0.$$

It starts positive and ends nonpositive, proving the unique-root claim and excluding all $d < d_*$. The endpoint $v = 1$ follows by continuity. Finally $\operatorname{kl}(r, q) \geq r\ell - \log 2$, so the binary exclusion includes every d excluded by the positive affine bound. □

Remark 4.2. If $q = 0$, the stopped density implies $\mathbb{P}^f(E) = 0$ and the direct variance inequalities give $\operatorname{gap}(\pi) \geq v/\kappa$. A numerical claim that q is small does not justify setting it equal to zero. The cap $H_f \leq 2nT$ may be combined with the affine entropy bound in a piecewise root test. The simpler theorem above does not need this cap.

4.1 A concrete, explicitly hypothetical budget

Take $T = 1$, $n = 10^8$, $C_T = 2$, $C_d \leq 1/2$, and use $v = 1/2$. Suppose that the actual stopped experiment has certified $q \leq e^{-n/16}$, $a = 1/16$, $B_0 = 2$, $B_1 = 1$. These are hypothetical interface constants, not constants extracted from a finite SK simulation. Since $\log 2 < 1$,

$$\frac{H_0}{n} = \frac{1}{512} + \frac{33}{160000}, \qquad \frac{H_1}{n} = \frac{33}{320000}, \qquad \kappa = \frac{5}{2}.$$

Formula (12) gives the rigorously rational lower bound

$$\operatorname{gap}(\pi) \geq \frac{\frac{1}{2} - 16(\frac{1}{512} + \frac{33}{160000}) - \frac{16}{10^8}}{\frac{5}{2} + \frac{33}{20000}} > \frac{9}{50}. \tag{15}$$

Here $e^{-C_d} \geq 1/2$ follows, for example, from $e^{1/2} = \sum_{k\geq 0}(1/2)^k/k! < 2$. For the same drift input, using only $b^2 \leq 2b$ gives $H_0^{\mathrm{crude}}/n = a + B_0/\sqrt{n}$, whose constant term in the ordinary-to-square bad-event bound already exceeds one. That affine certificate is vacuous. This comparison shows a genuine possible improvement of a finite budget; it is not measured mixing performance.

5 Likelihood residuals that preserve the physical laws

An approximate mean vector or a positive finite covariance matrix does not by itself describe a probability law. A useful residual certificate instead controls an actual likelihood on the entire cube.

Lemma 5.1 (Likelihood oscillation)**.** *Let $\widehat{\mu} = e^r\mu/\mu(e^r)$, where μ is a full-support probability on a finite set and $\sup r - \inf r \le \omega < \infty$. Then*

$$\mathrm{TV}(\widehat{\mu}, \mu) \le \tanh(\omega/4). \tag{16}$$

This constant is sharp over all such pairs of laws. For any nonzero f, the same bound holds between the normalized laws $f^2\widehat{\mu}/\widehat{\mu}f^2$ and $f^2\mu/\mu f^2$.

Proof. Write $w = e^r$, $l = \min w$, $u = \max w$, $Z = \mu w$. If $l = u$ the laws agree. Otherwise the convex chord for $|w - Z|$ gives

$$\mathrm{TV} = \frac{\mu|w - Z|}{2Z} \le \frac{(u - Z)(Z - l)}{Z(u - l)} \le \frac{\sqrt{u} - \sqrt{l}}{\sqrt{u} + \sqrt{l}}.$$

The last maximum occurs at $Z = \sqrt{lu}$, as follows by writing its numerator divided by Z as $u + l - Z - ul/Z$. Since $u/l \le e^\omega$, (16) follows. A two-point law on weights l, u with mean $\sqrt{lu}$ attains equality. On the support of f^2, the second pair has exactly the same multiplicative weight w and possibly a smaller oscillation. Apply the first assertion there; full support of the tilted law outside this restricted set is not required. □

Proposition 5.2 (Heat-bath and drift residual transfer)**.** *On the cube, under the hypotheses of Lemma 5.1, write $R = \mathcal{D}_\mu(f)/\mu f^2$, $\widehat{R} = \mathcal{D}_{\widehat{\mu}}(f)/\widehat{\mu}f^2$. Then*

$$e^{-\omega}R \le \widehat{R} \le e^\omega R, \qquad e^{-\omega}\,\mathrm{gap}(\mu) \le \mathrm{gap}(\widehat{\mu}) \le e^\omega\,\mathrm{gap}(\mu). \tag{17}$$

For the ordinary and square-density mean differences a_f and $\widehat{a}_f$,

$$\|a_f - \widehat{a}_f\|/\sqrt{n} \le 4\tanh(\omega/4). \tag{18}$$

Consequently a certified approximate drift inequality

$$\|\widehat{a}_f\|/\sqrt{n} \le \widehat{a} + \frac{\widehat{B}_0 + \widehat{B}_1\widehat{R}}{\sqrt{n}}$$

implies the true inequality (9) with

$$a = \widehat{a} + 4\tanh(\omega/4), \quad B_0 = \widehat{B}_0, \quad B_1 = e^\omega\widehat{B}_1, \tag{19}$$

provided the approximate constants are nonnegative and the resulting $a < 2$. The likelihood oscillation must be uniform over the stopped paths for use in Proposition 3.2.

Proof. For each unordered flip edge $\{x, y\}$, the heat-bath conductance is

$$c_\mu(x, y) = \frac{\mu(x)\mu(y)}{\mu(x) + \mu(y)}, \qquad \mathcal{D}_\mu(f) = \sum_{\{x,y\}} c_\mu(x, y)(f(x) - f(y))^2.$$

The map $(s, t) \mapsto st/(s+t)$ is increasing and homogeneous of degree one. Since $l/Z \le \widehat{\mu}/\mu \le u/Z$, both Dirichlet energies are comparable with factors $l/Z, u/Z$. The same comparison holds for μf^2, and for $\mathrm{Var}_\mu f = \inf_c \mu(f - c)^2$. The normalization Z cancels from the quotients; their comparison factor is $u/l \le e^\omega$, not the unnecessarily squared factor obtained by separately bounding normalized densities by $e^{\pm\omega}$.

For any two cube laws λ, λ' and unit vector e, $|e^{\mathsf{T}}x| \le \sqrt{n}$ and hence $|(\lambda - \lambda')e^{\mathsf{T}}x| \le 2\sqrt{n}\,\mathrm{TV}(\lambda, \lambda')$. Apply Lemma 5.1 to the ordinary pair and the square-density pair, then use the triangle inequality. Finally substitute $\widehat{R} \le e^\omega R$ into the approximate bound. □

The gap comparison in (17) is the classical bounded-potential comparison specialized to heat-bath conductances. Its role here is to keep the same physical likelihood, conditional updates and square reweighting together. No assertion is made that a polynomial surrogate, Gaussian moment match, or a set of sampled log-likelihood residuals provides the required uniform ω.

5.1 An explicit two-point sharpness fixture

Let the likelihood weights be $1, s^2$, $s > 1$, and let their original probabilities be $s/(s+1), 1/(s+1)$. The normalizer is s. The new probabilities are $1/(s+1), s/(s+1)$ and $\mathrm{TV} = (s-1)/(s+1) = \tanh((2\log s)/4)$. Thus a small total-variation conclusion cannot be deduced by omitting the square root of the likelihood range. For a nonconstant f, the square-density law generally has a different normalizer; it must be recomputed, although the same range bound still applies.

6 Why simpler certificates fail

6.1 Identity covariance does not imply a uniform gap

For $n \geq 3$ and $0 \leq \theta < 1$, define

$$\mu_\theta(x) = 2^{-n}(1 + \theta \prod_i x_i).$$

This law has full support, mean zero and covariance I_n. Orthogonality of cube characters verifies all these assertions. For $f = x_1$, $\operatorname{Var} f = 1$, while the conditional mean of x_1 is $\theta \prod_{i>1} x_i$ and its conditional variance is $1 - \theta^2$. Only site one contributes to $\mathcal{D}(f)$, so $\operatorname{gap}(\mu_\theta) \leq 1 - \theta^2 \to 0$. This is not a counterexample to the SK statement: it is a counterexample to replacing its all-function stability argument by positivity or boundedness of the linear covariance matrix.

6.2 Rare under one law, typical under the square law

For an event A with $\mu(A) = q \in (0,1)$, set $f = (\mathbf{1}_A - q)/\sqrt{q(1-q)}$. Then $\mu f = 0$, $\mu f^2 = 1$, but $(f^2\mu)(A) = 1 - q$. This exact example concerns a single probability space, not a constructed bad path for the SK observation experiment. It explains why the ordinary estimate $\mathbb{P}(E) \ll 1$ cannot simply be inserted as an estimate of $\mathbb{P}^f(E)$: the entropy calculation is indispensable.

6.3 A finite covariance residual misses omitted functions

If a finite family $\phi_1, \ldots, \phi_m$ is used to form $G_{ab} = \operatorname{Cov}_\mu(\phi_a, \phi_b)$ and $K_{ab} = \sum_i \mathbb{E}_\mu[(\phi_a - P_i\phi_a)(\phi_b - P_i\phi_b)]$, then $K \succeq gG$ proves the Poincare inequality only on that span. On the nonconstant subspace, the infimum restricted to a smaller span is an *upper* bound for the full gap, not a lower bound. The parity fixture shows how well-conditioned linear data can hide a slow direction. A valid global certificate requires all functions, or an independently proved control of the omitted subspace.

7 What the upstream proof must still provide

The entire cached text was read, but reading is not independent verification of every random-matrix, Gaussian or concentration argument. The following map records how the local certificates attach to the actual source, rather than to an abstract covariance surrogate.

7.1 Matrix words and the stable field

The matrix module defines, for a bounded nonnegative diagonal A,

$$B_A = j\ \mathrm{Tr}(A)/n, \quad K_A(z;J) = (I - zAJ + z^2 B_A A)^{-1}, \quad S_A(z;J) = I + z^2 B_A A - zA^{1/2} J A^{1/2}.$$

The adaptive inverse is used only with an actual lower margin $S_A \succeq cI$. The identity $K_A A = A^{1/2} S_A^{-1} A^{1/2}$ avoids assuming a lower bound on the individual entries of A. The source truncates the functional calculus before Gaussian integration by parts, derives a finite loop recursion, then uses a multiscale parameter mesh for simultaneous diagonal and off-diagonal word estimates. Its deterministic-vector bilinear estimate is for each prescribed finite family, whereas its regularized log-determinant bound is simultaneous over all allowed diagonals and bounded finite-rank perturbations. These quantifiers cannot be exchanged.

With $m = \tanh y$ and $q_y = \|m\|^2/n$, the nonlinear field map is

$$F_h(y) = y - h + [j(1-q_y)I - J]m. \tag{20}$$

The required stable region is not a single computed root. It controls all sufficiently small residuals and all allowed diagonals near $\mathrm{diag}(1-m_i^2)$ through

$$I + A^{1/2}\left[j(1-q_y)I - J - \frac{2j}{n} m m^{\mathsf{T}}\right] A^{1/2} \succeq cI.$$

The source's scalar entropy module identifies its Gaussian equality curve. The stability module then uses planted Gaussian regression, conditional finite-rank Hessian positivity, empirical-law tail control, and a local-volume estimate at any approximate zero. A time mesh and Brownian modulus transfer this exclusion to the path. Finally, a partition-function size bias is undone to recover ordinary observation probabilities for typical disorder. A finite positive Hessian at one field does none of this.

7.2 Residual recipes, buffers, and ordinary means

The primary iteration is

$$m^0 = x, \quad m^\ell = \tanh h^\ell, \quad b_\ell = n^{-1}\sum_i (1 - (m_i^\ell)^2), \quad h^{\ell+1} = h^0 + Jm^\ell - j b_\ell m^{\ell-1}.$$

The source does not require its infinite-depth convergence. Fixed finite-depth cutoffs select two adjacent small residuals $R_\ell = m^{\ell-1} - m^\ell$. For $S_\ell = R_\ell^{\mathsf{T}} m^\ell/\sqrt{n}$, the exact telescoping identity is

$$\|R_\ell\|^2/n = b_\ell - b_{\ell-1} - 2S_\ell/\sqrt{n}.$$

Testing these residuals requires the source's ordinary and square-density weak residual rules, including the averaged Onsager correction. For example,

$$d_i(f^2) = 2f\, d_i f - 2x_i (d_i f)^2$$

has a second term which cannot be suppressed in a square-density calculation. In the implicit direction test, small-vector norm bounds alone are insufficient: the proof needs a bounded diagonal derivative ℓ^1 norm, obtained by trace cancellation for finite recipes with at most one inverse factor in each word.

There are three distinct residual neighborhoods $\Omega_\rho \subset \Omega_{3\rho/2} \subset \Omega_{2\rho}$. The actual implicit solution is retained on the largest one. Tests are supported on the smallest; all finitely many spin flips used in their proofs remain inside the buffer once n exceeds a depth-dependent threshold. An arbitrary extension outside the buffer is harmless only with this support assertion. The susceptibility-average test is controlled before the direction test, allowing ρ to be chosen before the finite selection depth. This is what supports the unnormalized ordinary-mean bound used in (8).

7.3 Terminal all-function inequality and ordered choices

At a large fixed observation time the source proves its terminal $C_T = 2$ inequality from a signed two-spin curvature estimate. It keeps $U_{ij} = \tanh J_{ij}$ and the entrywise square matrix $Q_{ij} = U_{ij}^2$ separate. The first-order mixed terms cancel, leaving a fourth-power row error. Small principal-submatrix bounds handle the few weak-field sites, while the conditional variances on the other sites are small. An eigenfunction argument for the full heat-bath operator then converts the curvature bound into a gap. This is stronger than testing linear functions and is imported intact here.

The proof order remains: fix β and a suitable finite T; fix the stability and directional constants; choose a sufficiently small fixed δ; choose the resulting finite recipe depth and word length; then take n sufficiently large. The ordinary failure probability and the square-law failure probability remain different quantities. Theorem 4.1 refines the last transfer, not the preceding order. In particular the $C_\delta/\sqrt{n}$ term cannot be declared negligible at a numerically selected dimension without an actual bound on C_δ and the corresponding dimension threshold.

Source module	Scientific role	Status in this report
Introduction, observation	Model, clock, stopping and change of law	Fully read; transfer rederived and refined.
Matrices	Single-inverse words, adaptive diagnostics, determinant	Fully read; uniform probabilistic conclusions imported.
Entropy	Scalar inequality and Gaussian equality curve	Fully read; not replaced by finite moments.
Stability	Conditional Hessian, local volume, planting and path	Fully read; global exclusion and probability bounds imported.
Residuals	Discrete derivatives, buffered implicit recipes, traces	Fully read; no independent all-recipe certification asserted.
Consequences	Selection, directional estimate, both mean bounds	Fully read; stronger ordinary mean used explicitly.
Terminal	Two-spin curvature and large-field all-function gap	Fully read; retained as an input.

No correction of a source formula is asserted. The rate, conditional variance, square-density normalization, adaptive inverse gates and Onsager terms just listed are valid details retained, not defects.

8 Connection with original Ram proofs

The fresh private reading was targeted, not a whole-corpus survey. The original posterior-moment derivation in TR047 [6], lines 777–888, explicitly distinguishes moments under the actual posterior from moments under an approximation, and writes the error as change of law plus polynomial remainder. It also distinguishes the covariance of one common polynomial from separately truncated first and second moments. The former requires moments through twice the polynomial degree; the latter can be indefinite.

For completeness, its elementary covariance remainder statement follows directly: if $F = T + r$, $V_T = \mathbb{E}\|T - \mathbb{E}T\|^2$, and $\mathbb{E}\|r\|^2 \le \varepsilon^2$, expand the covariance into the covariance of T, the covariance of r, and two cross terms. Cauchy–Schwarz bounds the latter in operator norm by $\sqrt{V_T}\varepsilon$ each, and the former by ε^2. Thus the error is at most $2\sqrt{V_T}\varepsilon + \varepsilon^2$. This estimate concerns a specified common probability law. It does not compare π_t and $f^2\pi_t/N_t$, nor does it control all test functions. The likelihood formulation in Section 5 supplies a different, law-level sufficient condition suited to that distinction.

The original physical-metric chapter [7], freshly read in full, explains why a coefficient norm must be transported by the true Gram matrix and why a projected residual omits an orthogonal

component. For $u = Jc$ in a finite synthesis space, the physical norm is c^*J^*Jc, not c^*c unless J is an isometry. Its residual decomposition and its example of an entrywise-small but relatively unbounded Gram error support the warning in Section 6. They do not produce the missing all-function Poincare estimate.

Selected passages of the later organizing manuscript TR-core-pat-16 were also read as context [8]. They were not used as a substitute for the original proofs above. The catalogue lead TR-core-pat-5 was structurally located but not literally read or used as mathematical evidence. None of these private works, titles, or passages was sent to an online search.

9 Current public literature and status

The primary literature refresh on October 9, 2026 read the following abstracts and version metadata, not their full external proofs. Boban, Li and Oveis Gharan's September 11 preprint [2] reports polynomial-time Glauber mixing up to $\beta \le 1/2 + \varepsilon$ for an absolute $\varepsilon > 0$, using rank-one-perturbed trickledown. It is not a public all-$\beta < 1$ theorem on the strength of that abstract.

Wang's August 28 revised preprint [3] reports worst-start $O_\beta(n \log(n/\varepsilon))$ mixing for fixed $\beta < 1/2$, uniformly over external fields. Its stated route uses an integrated curvature criterion, a two-spin estimate, and a localization upgrade. This is a different range and stronger mixing-time conclusion than a bare Poincare assertion; the clocks and hypotheses must still be matched.

The localization framework of Chen–Eldan [4] and the information-theoretic viewpoint of El Alaoui–Montanari [5] give the classical context for martingales of measures and observation laws. The present sharpening does not claim to originate those ideas, binary relative-entropy contraction, bounded-potential comparison, or the affine-envelope calculation. No additional external theorem is imported to bypass the source's full-phase proof obligations.

10 Evidence, scope, and reproducibility

All 13 cached public text files were literally read in bounded, untruncated displays: 4,922 lines in total, comprising the eight scientific modules, the main file, bibliography, style, figure code and repository README. Their full-file line endpoints are

module	intro	observation	matrices	entropy	stability	residuals	consequences	terminal
last line	254	212	870	337	1207	1069	243	433.

The other endpoints are 26, 161, 42, 50 and 18, respectively. Thus there are no unread complements within these 13 text files. This does not mean that their cited literature, generated source PDF, or any external proof was read merely because it was referenced. The diagram's TeX code was read; this statement is not a separate visual audit of the upstream PDF.

Fresh private literal reading totals 542 distinct lines: TR047 777–888 (112 lines); the physical-metric chapter 1–210; TR-core-pat-16 99–180, 216–328 and 915–939 (220 lines). Only the first two are counted as original-proof reading here (322 lines). The rest is secondary organizational context. The per-file ledger preserves before/after SHA-256 values, every actual display interval, its merged union, and exact unread complements. Snapshot authentication against the captured repository commits is provenance, not a live-branch assertion or a mathematical certificate.

The owned script `M429-exact-checks.py` uses Python rational arithmetic and rational logarithm enclosures. For $1 \le x \le 2$ it uses $z = (x-1)/(x+1)$ and

$$\log x = 2 \sum_{k=0}^{m-1} \frac{z^{2k+1}}{2k+1} + R_m, \qquad 0 \le R_m \le \frac{2z^{2m+1}}{(2m+1)(1-z^2)}.$$

Positive rational inputs are scaled by powers of two before this formula is applied. Endpoint interval signs, rather than rounded decimal logarithms, validate the root brackets.

Owned check family	Cases	Assertions
Bounded-drift envelope	1200	2400
Sharp moment mixtures	272	544
Hypothetical transfer budget	1	3
Binary-entropy root brackets	9	315
Likelihood TV sharpness	4	8
Coherent likelihood and square law	80	960
Square-event obstruction	30	60
Total	1596	4290

A case is one parameter instance; a case may contain several asserted equalities or inequalities. These checks test local algebra, endpoint conventions and numerical certification logic. They do not sample Gaussian disorder, certify a rare-path exponent, audit infinite recipe families, or establish a large-n SK gap. Independent reviews and their test counts are not included in the owned totals.

11 Advantages, costs, and further work

Advantages. The improved entropy budget uses a stronger conclusion already proved inside the source, reduces its leading tolerance dependence from linear to quadratic, and yields an explicit all-d lower-bound calculation. The binary root retains information discarded by the $\log 2$ bound. The residual theorem preserves both the physical update law and its square tilt. All formulas keep the dimension, clock, path law and conditional hypotheses visible.

Costs. The constants C_δ, the depth-dependent dimension threshold, the ordinary failure exponent and the terminal time are not quantified by this report. Computing all 2^n likelihood values is generally infeasible; a uniform oscillation certificate needs analytic structure or a separate validated method. The total-variation drift bound uses only the spin diameter and may be conservative. A root solve improves an established error budget, not the underlying probability estimates.

Feasible extensions. Time-dependent certified drift constants can be retained in the entropy integral instead of replacing them by suprema: its constant term becomes $\frac{1}{2}\int_0^T [na(t)^2 + \sqrt{n}(2+a(t))B_0(t)]\,dt$, and its coefficient of d becomes $\frac{\sqrt{n}}{2}\int_0^T (2+a(t))B_1(t)\,dt$. The proof is identical using deterministic nonnegative coefficients and (6). This does not allow data-dependent coefficients without controlling their correlations. One could also retain a sharper measured bound on $\mathbb{E}^f \mathbf{1}_{\{t<\tau\}} R_t$ when it is proved uniformly in f. An important separate research problem is quantitative control of C_δ and of the stability/recipe thresholds, which would make finite-dimensional use less hypothetical.

Assessment. Applicability L3 concerns the relevance of the Ram moment and physical metric viewpoints. The independent contribution rating is C2: the report provides useful, source-specific quantitative refinements, but the tools and comparison principles are classical and the principal high-temperature mechanisms remain those of the source. Neither positive finite Gram matrices nor these tests replace the all-function and all-path quantifiers.

References

[1] OpenAI. *A spectral gap throughout the high-temperature Sherrington–Kirkpatrick phase.* September 24, 2026. Pinned local manuscript, eight scientific modules and supporting text. Public manuscript directory. The repository URL is a locator; byte identity is recorded separately.

[2] M. Boban, A. Li and S. Oveis Gharan. *Rank-1-perturbed trickledown theorems: Mixing time of Glauber dynamics for the Sherrington–Kirkpatrick model up to* $\beta \leq 1/2 + \varepsilon$. September 11, 2026. arXiv:2609.13138v1.

[3] S. Wang. *Optimal Mixing of Glauber Dynamics for the Sherrington–Kirkpatrick Model at* $\beta < 1/2$. Revised August 28, 2026. arXiv:2608.22159v2.

[4] Y. Chen and R. Eldan. *Localization Schemes: A Framework for Proving Mixing Bounds for Markov Chains*. arXiv:2203.04163v2, June 6, 2022. Duke Mathematical Journal 174 (2025), 1431–1510 (journal bibliographic data as cited in the assigned source).

[5] A. El Alaoui and A. Montanari. *An Information-Theoretic View of Stochastic Localization*. arXiv:2109.00709v2, September 9, 2021.

[6] M. Subbarao. Original Ram Foundation manuscript TR047, revised version. Private local corpus, posterior-moment propagation derivation, lines 777–888. Exact path and hash retained in the private read ledger.

[7] M. Subbarao. Ram Calculus, Volume 21, physical-metrics chapter. Private original local text, lines 1–210; physical Gram norms, omitted residuals and finite-measurement boundaries.

[8] M. Subbarao. TR-core-pat-16. Private organizing manuscript; selected probability-law and moment-consistency passages only. Used as secondary context, not as an independent SK theorem.